

Adalat AI: Concepts & Examples Guide

Mastering Courtroom Advocacy and Multi-Agent AI Software with Real-World Scenarios.
Designed for law students, legal educators, and software engineers.

WHAT THIS GUIDE COVERS

Parts 1 to 3: The Core Problem, Courtroom Roles & 5 Phases of Trial with Dialogue Examples.

Part 4: The Golden Rule of Leading Questions & 5 Objection Types with Real Scenarios.

Part 5: Pakistani Statutes (QSO, PPC, CrPC, Constitution) Explained in Simple Terms.

Parts 6 to 9: Multi-Agent AI Architecture, Hybrid RAG Search, ReAct Judge & Dual Memory.

Part 1

What Problem Does Adalat AI Solve?

Law students cannot become effective advocates just by reading textbook rules. To succeed in court, a lawyer must learn to think on their feet, argue out loud while standing up, and handle interruptions from an active judge. The standard way to practice this is a **moot court** — a pretend trial where students argue fictional cases.

REAL-WORLD MOOT COURT CHALLENGE VS. ADALAT AI

Traditional Practice Struggle: Organizing a pretend trial requires 4 to 5 people in a room: a student lawyer, an opposing lawyer, a judge, and 2 witnesses. Coordinating everyone's schedule is so difficult that students rarely get more than 1 or 2 practice sessions per semester.

The Adalat AI Solution: Adalat AI replaces all human participants with specialized AI agents. A student puts on a headset at 11 PM, speaks into a microphone, and argues against an AI Opposing Lawyer and AI Judge. Each character speaks back in a distinct voice, giving the student unlimited practice.

The Core Danger of AI in Law: An AI tool that confidently quotes a fake law is worse than no tool at all. If a student memorizes a hallucinated section number, they might repeat it in front of a real High Court judge. Therefore, Adalat AI is built around **provable legal honesty**: it only allows citations that actually exist in verified law.

Part 2

Courtroom Roles & Terminology

Before understanding courtroom software, you must understand who sits in the room and what they do:

Term / Role	Plain-English Meaning	Real-World Example
The Bench	The Judge presiding over the court.	Judge Ahmad sitting at the elevated desk making legal rulings.
Counsel	The lawyer arguing a case (You!).	Learned counsel addressing the judge: 'My Lord, I represent the victim.'

Term / Role	Plain-English Meaning	Real-World Example
Opposing Counsel	The lawyer fighting against you.	The prosecutor trying to prove your client is guilty.
Witness	Someone giving sworn testimony under oath.	A bank manager testifying about security camera footage.
Petitioner	The person/side who filed the case in court.	A landlord filing a petition to evict a non-paying tenant.
Respondent	The side defending against the case.	The tenant responding to defend their lease agreement.

Part 3

The 5 Phases of a Court Trial

A trial proceeds in a strict 5-stage order modeled by Adalat AI:

- **1. Opening Statements:** Each lawyer gives a brief preview of what the case is about and what outcome they request.
- **2. Examination-in-Chief (Direct Examination):** You question *your own witness* to bring out facts supporting your side.
- **3. Cross-Examination:** You question *the other side's witness* to expose lies, contradictions, or memory gaps.
- **4. Closing Arguments:** Each lawyer summarizes the whole evidence and asks the judge for a final decision.
- **5. Verdict & Marking:** The judge renders the final verdict and evaluates the student's legal performance.

PRACTICAL DIALOGUE EXAMPLES FOR THE 5 PHASES

Phase 1 (Opening): *'My Lord, we will prove that the defendant dishonestly issued a cheque of Rs. 500,000 without sufficient bank balance.'*

Phase 2 (Examination-in-chief): *'Mr. Witness, what did you see when you entered the warehouse at 9 PM?'*

Phase 3 (Cross-examination): *'Isn't it true you were standing 100 meters away in the dark when you claimed to see the face?'*

Phase 4 (Closing): *'The evidence shows no common intention under PPC 34; therefore, my client must be acquitted.'*

Phase 5 (Verdict): *'The court finds the accused guilty under Section 489-F. Student score: 88/100.'*

Part 4

The Golden Rule: Leading Questions & Objections

The central legal principle behind courtroom software is the rule governing **Leading Questions**. A leading question is one that puts the desired answer directly into the witness's mouth.

THE SIMPLE TEST FOR LEADING QUESTIONS

BAD (Leading Question in Direct Exam): 'You saw Bilal break the shop lock with an iron rod, didn't you?'
→ *Why it is illegal:* You fed the answer to your witness. All they have to do is nod or say 'yes'.

GOOD (Open Question in Direct Exam): 'What did you observe when you looked toward the shop front?'
→ *Why it is allowed:* The witness gives their own story in their own words.

Rule Summary: Questions starting with **What, Where, When, Who, Why, How,** or **Describe** are almost NEVER leading.

Common Objection Grounds & Real Scenarios

When a lawyer asks an improper question, the opponent shouts '**Objection, My Lord!**' followed by a ground:

Ground	Plain Meaning	Real Courtroom Example
Leading Question	Feeding answers to your own witness during examination-in-chief.	'You were at the market at 8 PM, right?'
Hearsay	Repeating what someone else said outside court instead of direct knowledge.	'My brother told me he saw Tariq sign the fake document.'
Irrelevant	Question has zero connection to the facts of the case.	Asking a robbery witness: 'What did you eat for breakfast today?'
Argumentative	Badgering, insulting, or arguing with the witness.	'Are you expecting this court to believe such an obvious lie?'
Assumes Unproved Facts	Treating an unproven accusation as an established fact.	'When did you stop stealing money from the cash register?'

JUDGE'S RULING OPTIONS

1. Sustains Objection: The Judge agrees the question breaks a rule. '*Sustained! Counsel, rephrase your question.*' The witness is **strictly forbidden** from answering.

2. Overrules Objection: The Judge disagrees with the objection. '*Overruled! The question is proper. The witness may answer.*'

Part 5

Pakistani Statutes Explained in Simple English

Adalat AI operates on 53 core provisions across 4 major Pakistani statutes:

Statute	Full Name	What It Controls in Simple Terms
QSO 1984	Qanun-e-Shahadat Order, 1984	The Law of Evidence: Rules on what counts as proof, how witnesses are questioned, and what documents are valid.
PPC 1860	Pakistan Penal Code, 1860	Criminal Code: Defines criminal offenses (murder, theft, fraud, rape) and their penalties.

Statute	Full Name	What It Controls in Simple Terms
CrPC 1898	Code of Criminal Procedure, 1898	Criminal Procedure: Step-by-step rules on how police investigate, how FIRs are filed, how bail is granted, and how trials run.
Constitution 1973	Constitution of Pakistan, 1973	Supreme Law: Fundamental human rights, fair trial guarantees, protection against double jeopardy, and High Court writs.

Key Provisions & Real-World Examples

- **QSO Article 71 (Direct Evidence):** Oral evidence must be direct. Example: An eyewitness testifying '*I saw the crash*' is admissible; testifying '*A bystander told me there was a crash*' is inadmissible hearsay.
- **PPC Section 302 (Punishment for Qatl-i-amd):** Penalty for intentional murder. Example: Intentionally shooting a rival carries death (Qisas) or life imprisonment (Tazir).
- **PPC Section 489-F (Dishonest Cheque):** Issuing a cheque without funds. Example: Writing a business cheque for Rs. 1,000,000 knowing your account has Rs. 0 is a crime punishable by up to 3 years prison.
- **CrPC Section 154 (First Information Report / FIR):** The legal starting point of any major criminal investigation recorded by police.
- **Constitution Article 10A (Right to Fair Trial):** Guarantees every citizen a fair, open hearing before an independent court.
- **Constitution Article 13 (Double Jeopardy):** Protects a citizen from being tried or punished twice for the exact same offense.

Part 6

How the AI Software Works (Technology Architecture)

Adalat AI is built as a modular multi-agent system divided into 3 distinct software layers:



WHY SPLIT THE SOFTWARE LIKE THIS?

1. Total Control over AI Logic: All AI reasoning lives in a single Python folder. Developers never have to hunt through web code to find out why the AI judge made a decision.

2. Security & Password Protection: The browser never touches OpenAI API keys. A user inspecting web developer tools cannot steal the API key because it never leaves the backend server.

Part 7

How the AI Finds Laws: Hybrid RAG & Reranking

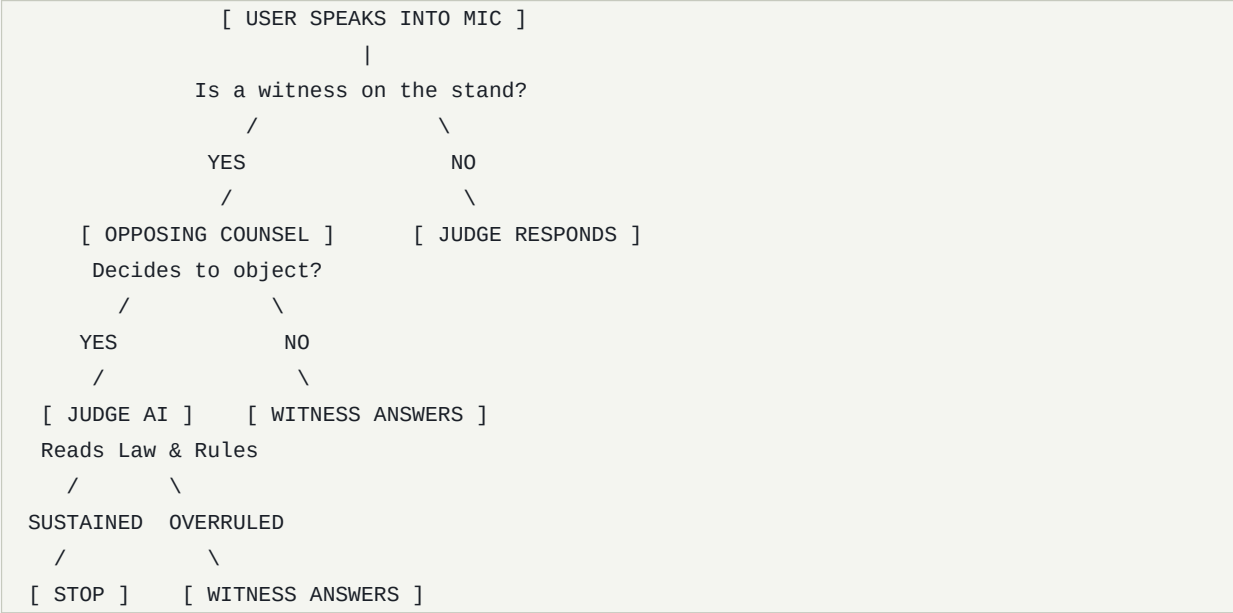
If you ask a normal AI model to cite a law, it will confidently invent fake section numbers. Adalat AI solves this using **RAG (Retrieval-Augmented Generation)** — searching a database of 53 real laws first, then forcing the AI to answer strictly using the retrieved text.

Search Type	How It Works	Real Example Query & Match
Keyword Match (BM25)	Matches exact words and numbers.	Query: 'Section 302 PPC' → Matches exact Section 302 text.
Meaning Match (Vector)	Converts sentences into mathematical vectors of meaning.	Query: 'The witness is just repeating rumors.' → Matches QSO Article 71 (Hearsay) even with zero word overlap.
Reciprocal Rank Fusion (RRF)	Combines search ranks fairly.	Merges Keyword rank #1 and Vector rank #2 into a single unified shortlist.
AI Reranking	A second AI re-reads the shortlist.	Re-orders the top 3 provisions to put the exact governing law in position #1 (100% accuracy).

Part 8

Multi-Agent Graph & Structural Rule Enforcement

The courtroom is not one single AI pretending to be multiple characters. It is a **flowchart** of separate specialized AI agents (State Graph):



STRUCTURAL IMPOSSIBILITY OF LEAKS

Notice what happens when the judge **sustains** an objection: the flowchart arrow goes directly to **STOP**. There is no arrow connecting Sustained to the Witness. The witness does not stay quiet because the AI was politely told not to speak — it stays quiet because **there is no code path in the flowchart that allows it to speak**. Zero leaks in 32 test scenarios!

Part 9

Smart Cost Saving, Dual Memory & Student Grading

Model Cascading (Cheap Filter + Smart Decider)

Checking every single question with an expensive AI model is wasteful. Adalat AI uses a **cascade**: A fast, cheap AI model inspects every question first. If it sees nothing wrong, the turn ends cheaply (\$0.0020). If the question looks suspicious, it passes it to the expensive smart model to make the final ruling. This cuts average turn costs to \$0.0095 without losing any accuracy.

Dual-Layer Memory (Catching Contradictions)

- **Working Memory (Free):** Remembers exact word-for-word exchanges in the current trial phase.
- **Long-Term Memory (~4 AI calls/session):** Maintains a structured list of claims the student committed to.
- **Real Example:** Opposing counsel catching a student in closing argument: *'In your opening, you stated your client was out of town. Now you say he was at the scene! Which is it?'*

Student Marking & Verification Guardrails

At the end of a trial, the system pulls every statutory reference out of what the student said and checks if each law exists in the database. If a student cites a fake law (e.g. 'Section 999 PPC'), they are penalized. The student receives 5 scores out of 100: **Legal Reasoning, Persuasiveness, Procedural Command, Factual Command, and Overall Score.**

SUMMARY OF ADALAT AI PERFORMANCE METRICS

- **Legal Retrieval Accuracy:** 100% (after Reranking).
- **Objection Accuracy:** 100% Precision, Recall, and Specificity across 32 test scenarios.
- **Witness Leaks on Sustained Objections:** 0 occurrences.
- **Hack Attack Protection:** 0 out of 36 prompt injection attacks succeeded.
- **Average Cost per Spoken Turn:** \$0.0095.
- **Verified Laws Corpus:** 53 provisions fully verified across 4 Pakistani statutes.

Guide compiled from the Adalat AI source code. Designed to provide a complete understanding of courtroom trial procedures, evidence rules, and multi-agent AI engineering.